

BRONTE 11 NNE, TX, USA

WMO: 747370

Lat: 32.0408N

Lon: 100.2495W

Elev: 609

StdP: 94.22

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 03072

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.4	-5.3	-14.6	1.1	-1.8	-12.1	1.4	2.3	6.2	12.5	5.6	13.4	1.2	N/A	0.383

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	38.5	20.5	37.2	20.7	36.0	20.7	23.8	31.7	23.2	31.4	22.7	31.0	3.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.9	17.8	26.0	21.1	17.0	25.7	20.4	16.3	25.6	74.3	31.4	72.0	31.5	69.9	31.0	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.5	4.8	4.3														
			DB	-11.8	40.3	2.2	1.6	-13.4	41.5	-14.7	42.5	-15.9	43.4	-17.5	44.6	
			WB	-12.5	24.2	2.3	0.8	-14.1	24.8	-15.4	25.2	-16.7	25.7	-18.3	26.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	6.3	9.3	14.0	18.2	22.2	26.8	27.7	27.8	24.1	17.9	12.0	7.4
	DBStd	8.81	4.97	6.02	5.29	4.65	4.24	2.69	2.15	2.21	3.17	4.81	5.05	5.06
	HDD10.0	397	134	78	24	4	0	0	0	0	0	6	37	113
	HDD18.3	1455	373	256	149	58	15	0	0	0	2	66	194	341
	CDD10.0	3262	20	58	148	248	377	505	549	552	424	251	98	31
	CDD18.3	1279	0	2	15	53	133	255	291	294	177	53	5	1
	CDH23.3	15774	19	87	268	719	1531	3085	3723	3716	1856	649	99	21
	CDH26.7	8159	2	21	73	280	721	1668	2121	2152	886	217	17	2
Wind	WSAvg	1.9	1.7	2.0	2.3	2.5	2.3	2.2	1.8	1.6	1.6	1.7	1.6	1.6
Precipitation	PrecAvg	595	24	32	40	48	79	77	44	69	61	55	39	24
	PrecMax	865	68	121	123	98	189	187	118	196	171	145	120	144
	PrecMin	310	1	0	1	4	35	4	3	0	3	3	1	0
	PrecStd	161	19	33	28	32	42	56	34	58	49	40	34	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.8	30.1	31.3	35.2	38.1	40.0	39.6	39.7	37.2	33.1	29.0	26.1
		MCWB	12.5	14.1	15.2	15.8	18.8	20.4	20.3	20.6	19.9	19.3	15.9	13.5
	2%	DB	22.6	26.0	28.8	32.4	35.7	37.9	37.8	38.3	35.1	31.2	26.0	22.5
		MCWB	11.1	13.6	14.4	16.2	19.1	20.6	20.6	20.9	20.2	18.8	15.4	12.2
	5%	DB	19.5	23.2	26.4	30.0	33.0	36.1	36.6	37.0	33.6	29.2	23.9	19.9
		MCWB	9.9	12.7	14.4	16.4	19.0	20.8	20.9	21.2	20.1	18.1	14.7	11.1
	10%	DB	16.5	20.3	24.0	27.6	30.7	34.4	35.3	35.6	31.8	27.2	21.6	17.0
		MCWB	8.7	11.6	14.2	16.1	18.8	21.1	20.9	21.0	19.9	17.2	13.8	10.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.1	17.3	18.7	20.4	22.9	24.5	24.6	24.2	23.8	21.8	19.1	16.6
		MCDB	20.7	23.1	25.8	28.1	30.8	32.3	32.0	32.6	29.9	28.8	25.3	21.1
	2%	WB	12.5	15.9	17.7	19.6	22.2	23.6	23.7	23.4	22.7	21.0	17.7	14.5
		MCDB	18.6	21.8	24.4	26.7	29.5	31.6	31.2	32.8	29.2	27.6	22.4	18.9
	5%	WB	10.9	14.8	16.7	18.8	21.5	22.9	23.1	22.8	22.1	20.1	16.3	13.0
		MCDB	17.9	20.7	22.4	25.6	28.5	31.1	31.0	32.6	28.6	26.1	20.9	17.8
	10%	WB	9.3	12.8	15.5	17.9	20.7	22.3	22.5	22.2	21.5	19.1	15.1	11.0
		MCDB	16.4	18.8	21.2	24.4	27.3	30.5	30.7	31.8	28.0	24.5	19.9	16.7
Mean Daily Temperature Range	5% DB	MDBR	15.4	15.7	15.6	15.1	13.5	13.2	13.7	13.9	13.0	15.0	15.4	14.7
		MCDBR	21.5	20.5	19.3	18.9	17.3	16.0	16.0	16.5	15.9	17.5	18.1	19.8
		MCWBR	11.1	10.2	9.2	8.1	6.1	4.6	4.8	4.6	5.0	7.8	9.2	10.8
	5% WB	MCDBR	17.8	17.1	14.6	14.1	13.3	13.7	13.0	13.9	12.4	12.8	13.2	14.9
		MCWBR	10.5	10.4	7.9	7.2	5.6	4.8	4.3	4.2	4.1	6.2	8.1	9.5
Clear-Sky Solar Irradiance	taub	0.286	0.300	0.318	0.363	0.382	0.400	0.413	0.398	0.372	0.349	0.316	0.296	
	taud	2.556	2.516	2.483	2.363	2.381	2.364	2.386	2.418	2.418	2.473	2.492	2.528	
	Ebn at Noon	949	964	965	927	906	884	871	882	895	895	900	915	
	Edh at Noon	82	94	105	123	122	124	121	115	111	97	87	80	
All-Sky Solar Radiation	RadAvg	3.27	3.98	5.16	6.28	6.57	7.17	6.96	6.40	5.34	4.31	3.43	2.89	
	RadStd	0.34	0.51	0.37	0.46	0.60	0.50	0.46	0.40	0.39	0.61	0.41	0.32	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (18 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140	+100

Nomenclature: See separate page